

IDX Exchange — Week 1 Deliverable

Housing Dataset: Feature Utility Assessment

This document summarizes an initial review of MLS dataset columns by predicted utility for training a close price prediction model. Columns are grouped by the kind of information they encode and their expected relationship to **ClosePrice** (the target variable). The goal is not to enumerate every field but to explain the reasoning behind each grouping so that feature selection decisions are documented and defensible.

Useful

These fields are expected to be directly informative about ClosePrice. They encode property characteristics, size, condition, carrying costs, and market context — the core inputs to any hedonic pricing model.

Feature Group	Rationale
Target Variable	ClosePrice is the prediction target. All other feature utility judgments are made relative to how informative they are about final sale price.
Amenity Flags	ViewYN, WaterfrontYN, BasementYN, PoolPrivateYN, NewConstructionYN, AttachedGarageYN. Binary presence/absence indicators for high-value amenities. Each signals a discrete price premium; collectively they capture qualitative differences not encoded in size or age alone.
Capacity Counts	BedroomsTotal, BathroomsTotalInteger, FireplacesTotal, GarageSpaces, ParkingTotal, MainLevelBedrooms, Stories, Levels. Integer counts of functional spaces. Strongly correlated with ClosePrice; bedroom and bathroom counts are among the most standard comparables in real estate valuation.
Size / Area	LivingArea, LotSizeSquareFeet, LotSizeArea, LotSizeAcres, BuildingAreaTotal. Overlapping measurements of interior and land footprint. Price scales with area; redundancy across columns allows imputation and cross-validation. LotSizeDimensions is string-formatted and would need parsing before use.
Property Type	PropertyType, BusinessType. Coarse categorical encoding of what kind of property is being sold. Essential for segmenting the model or as a feature, since price distributions differ substantially across residential, commercial, and land listings.
Listing Price	ListPrice is the seller's asking price at time of listing. Highly predictive of ClosePrice but risks data leakage if the model is meant to predict price independently; worth including with caution and awareness of its role.
Market Exposure	DaysOnMarket. A proxy for demand and desirability. Longer time on market typically correlates with lower relative close-to-list ratios; inversely related to buyer competition.
Property Age	YearBuilt. Age of structure. Interacts with condition and renovation history; newer construction generally commands a premium, though this is nonlinear.
HOA Costs	AssociationFee, AssociationFeeFrequency. Ongoing ownership costs that affect buyer affordability calculations and thereby effective demand. Relevant especially for condos and planned communities.
Tax Assessment	TaxAnnualAmount, TaxYear. Correlated with assessed value, which itself tracks market value over time. Useful as a secondary valuation signal, though lagged relative to current market conditions.

Market Context	MLSAreaMajor, MlsStatus. MLSAreaMajor provides a coarser geographic/market classification than city or ZIP, useful for capturing local market conditions without high cardinality. MlsStatus encodes the transaction stage and filters to appropriate listing cohorts.
Flooring / Original	Surface and condition-related fields. Assumed to encode material quality or renovation status; exact semantics depend on the MLS schema. Grouped here pending confirmation of encoding.

Semi-Useful

These fields carry real information but require additional processing, carry high cardinality, or serve primarily administrative or classificatory functions. They may be useful in specific modeling contexts but are not plug-and-play features.

<i>Feature Group</i>	<i>Rationale</i>
Listing IDs	ListingKey, ListingKeyNumeric, ListingId, StreetNumberNumeric. Unique row identifiers. Not predictive directly but needed for deduplication, joining, and tracking observations across tables.
Transaction Dates	ListingContractDate, PurchaseContractDate, CloseDate, ContractStatusChangeDate. Temporal markers of the deal lifecycle. Useful for extracting seasonality features (month, quarter) or computing time-to-contract spans, but secondary to structural features as direct inputs.
Fine-Grained Geography	Latitude, Longitude, City, SubdivisionName, CountyOrParish, BuyerAgentAOR, ListAgentAOR, BuyerOfficeAOR. Granular location identifiers. Latitude and longitude enable spatial modeling; others are high-cardinality categoricals that may generalize poorly without grouping or embedding.
Property Subtype	PropertySubType. More granular than PropertyType. Useful in combination with it but may introduce sparsity if many subtypes have few observations.
Office / Agent Names	ListOfficeName, BuyerOfficeName, CoListOfficeName. Brokerage-level identifiers. Could proxy for market positioning or agent quality effects, but are not generalizable across datasets or time and would require careful encoding.
School Districts	ElementarySchool, MiddleOrJuniorSchool, HighSchool, and associated district fields. School quality is a known price driver in residential markets, but these fields are high-cardinality strings. Useful if mapped to external quality indices; noisy as raw categoricals.
Builder Name	BuilderName. Relevant for new construction comparisons but sparse across the full dataset and not generalizable.

Not Useful

These fields are excluded from modeling. They are either constant across the dataset, personal identifiers with no predictive value, redundant with better-structured alternatives, or too sparse to be reliable in this geographic context.

<i>Feature Group</i>	<i>Rationale</i>
Constant State Fields	StateOrProvince, OriginatingSystemName, OriginatingSystemSubName. No variance within this California-sourced dataset. Encode no information useful for prediction.
Agent Contact Info	ListAgentEmail, ListAgentFirstName, ListAgentLastName, CoListAgentFirstName, CoListAgentLastName, BuyerAgentFirstName, BuyerAgentLastName,

	BuyerAgentMlsId, CoBuyerAgentFirstName. Personal identifiers. No predictive value; including them risks memorization of agent-specific artifacts rather than property characteristics.
Unparsed Address	UnparsedAddress, PostalCode. Address string is redundant given structured geographic fields; ZIP code is useful in principle but is already proxied by MLSAreaMajor and latitude/longitude.
Above/Below Grade Area	AboveGradeFinishedArea, BelowGradeFinishedArea. Mostly missing in California listings, where split-level and basement construction is uncommon. Useful only if present, which is rare in this market.

Flagged for Review

Fields whose utility could not be determined from schema information alone. Requires inspection of actual values.

<i>Field</i>	<i>Open Question</i>
CoveredSpaces	Unclear whether this is redundant with GarageSpaces and ParkingTotal or captures a distinct category (carports, etc.). Requires inspection of populated values to determine if it adds information beyond existing capacity counts.

Note: AOR = Association of REALTORS®. MLS = Multiple Listing Service. Feature groupings reflect the California residential market context of this dataset. Assessments may differ for datasets covering other geographies.